

38. DMA Controller for the Ethernet Controller (EDMACa)

38.1 Overview

This MCU has three channels for the ethernet controller direct memory access controller (EDMAC): two channels for the ethernet controller (ETHERC) and one channel for the PTP controller (EPTPC). EDMAC0 and EDMAC1 control data transmission and reception for ETHERC0 and ETHERC1, respectively. The PTPEDMAC controls data transmission and reception for ETHERC0 and ETHERC1 according to the EPTPC settings. In this chapter, channel numbers are indicated as n ($n = 0, 1$).

The EDMAC controls most of the transmit/receive buffer management for communications. This reduces the load on the CPU and allows efficient data transmission and reception. The data transfers are controlled according to the information (referred to as descriptors) on the memory.

Table 38.1 lists the EDMAC specifications. Figure 38.1 shows the EDMAC configuration, and Figure 38.2 shows the configuration of descriptors and transmit/receive buffers on the memory.

Table 38.1 EDMAC Specifications

Item	Description
Data transmission and reception	- Controls data transmission and reception according to descriptors - Supports single buffer frame transmission and reception (1 buffer per frame), and multi-buffer frame transmission and reception (multiple buffers per frame)
Functions	- Minimizes system bus occupation time using block transfer (32-byte units) - Writes back the transmit/receive frame state to descriptors - Inserts padding in receive data
Low power consumption function	The EDMAC can be set to the module-stop state to reduce power consumption.

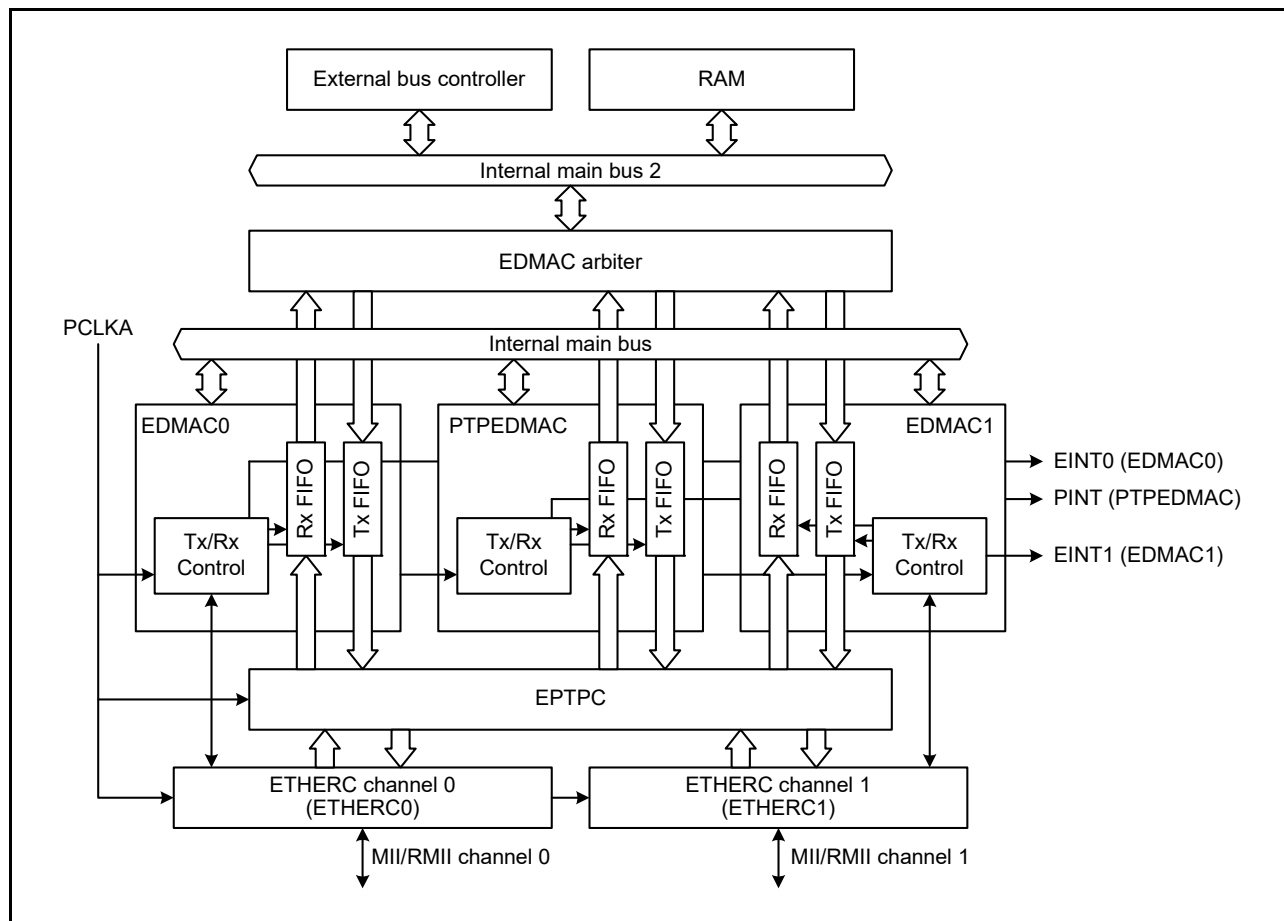


Figure 38.1 EDMAC Configuration

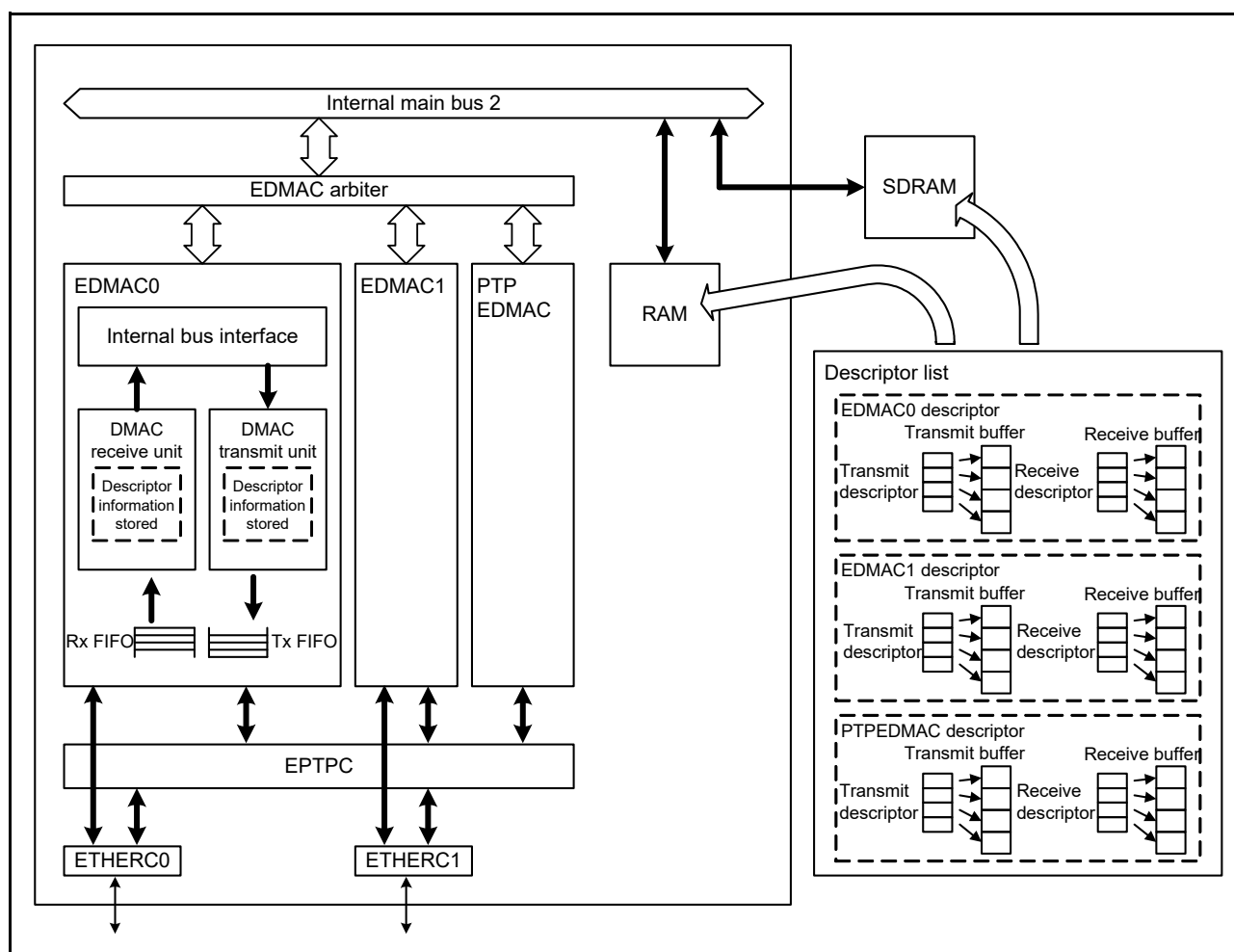


Figure 38.2 Configuration of Descriptors and Transmit/Receive Buffers on the Memory

38.2 Register Descriptions

38.2.1 EDMAC Mode Register (EDMR)

Address(es): EDMAC0.EDMR 000C 0000h, EDMAC1.EDMR 000C 0200h, PTPEDMAC.EDMR 000C 0400h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	DE	DL[1:0]	—	—	—	—	SWR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	SWR	Software Reset	When 1 is written, the corresponding channels of the EDMAC and ETHERC are reset. Note that for the PTPEDMAC, the ETHERC are not reset. Registers TDLAR, RDLAR, RMFCR, TFUCR, and RFOCR are not reset with this bit. The read value is 0.	R/W
b3 to b1	—	Reserved	The read value is 0. The write value should be 0.	R/W
b5, b4	DL[1:0]	Transmit/Receive Descriptor Length*1	b5 b4 0 0: 16 bytes 0 1: 32 bytes 1 0: 64 bytes 1 1: 16 bytes	R/W
b6	DE	Big Endian Mode/Little Endian Mode *1, *2	0: Big endian mode 1: Little endian mode	R/W
b31 to b7	—	Reserved	The read value is 0. The write value should be 0.	R/W

Note 1. Do not rewrite these bits while the transmit or receive function is enabled.

Note 2. This setting applies to data for the transmit/receive buffer. It does not apply to transmit/receive descriptors and registers.

The EDMR register controls EDMAC operation.

Set the EDMR register during the initialization process after a reset. If this register is rewritten during data transmission or reception, abnormal data may be sent on the line.

When rewriting this register outside of the initialization process, set the SWR bit to 1 to reset the EDMAC and ETHERC, and then set this register again. It takes 64 cycles of the peripheral module clock (PCLKA) to initialize the ETHERC and EDMAC. Complete the initialization process before accessing registers in the ETHERC and EDMAC. Refer to section 36.5.4, Procedure to Reset Ethernet Controller for the software reset procedure.

38.2.2 EDMAC Transmit Request Register (EDTRR)

Address(es): EDMAC0.EDTRR 000C 0008h, EDMAC1.EDTRR 000C 0208h, PTPEDMAC.EDTRR 000C 0408h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	TR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	TR	Transmit Request	When 1 is written, the EDMAC reads the corresponding descriptor and transmits frames where the TD0.TACT bit is 1. The TR bit becomes 0 after all the valid frames are transmitted. Writing 0 to this bit has no effect.	R/W
b31 to b1	—	Reserved	The read value is 0. The write value should be 0.	R/W

The EDTRR register controls EDMAC transmission.

After the EDMAC has transmitted one frame, it reads the next descriptor. When the TD0.TACT bit in the descriptor is 1, the EDMAC continues transmission. When the TD0.TACT bit is 0, the EDMAC sets the TR bit to 0 and stops transmission.

38.2.3 EDMAC Receive Request Register (EDRRR)

Address(es): EDMAC0.EDRRR 000C 0010h, EDMAC1.EDRRR 000C 0210h, PTPEDMAC.EDRRR 000C 0410h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	RR	Receive Request	0: Receive function is disabled. *1 1: Receive descriptor is read, and the receive function is enabled.	R/W
b31 to b1	—	Reserved	The read value is 0. The write value should be 0.	R/W

Note 1. If the receive function is disabled during frame reception, write-back to the receive descriptor is not performed successfully. Subsequent pointers for reading a receive descriptor become abnormal and the EDMAC cannot operate normally. In this case, to enable the EDMAC receive function again, execute a software reset by setting the EDMR.SWR bit to 1.

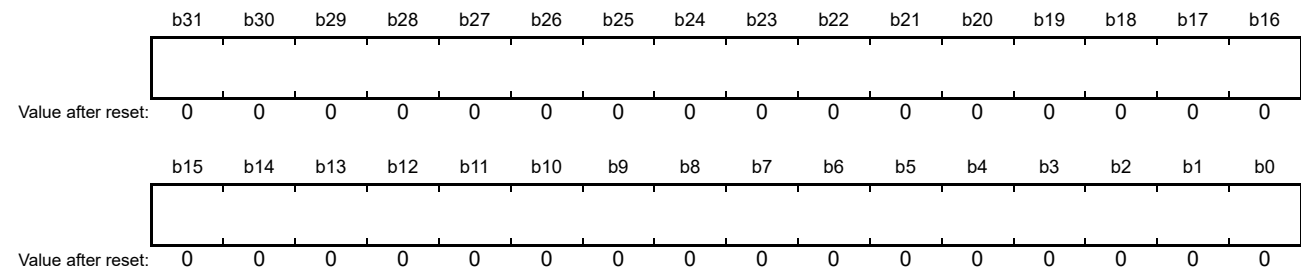
The EDRRR register controls EDMAC reception.

When the RR bit becomes 1, the EDMAC reads the receive descriptor. When the RD0.RACT bit is 1, the EDMAC waits for a receive request from the ETHERC.

When the EDMAC has received data for the receive buffer size, it reads the next descriptor and waits to receive a frame. If the RD0.RACT bit is 0, the EDMAC sets the RR bit to 0 and stops reception.

38.2.4 Transmit Descriptor List Start Address Register (TDLAR)

Address(es): EDMAC0.TDLAR 000C 0018h, EDMAC1.TDLAR 000C 0218h, PTPEDMAC.TDLAR 000C 0418h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	The start address of the transmit descriptor list is set. Set the start address according to the descriptor length selected by the EDMR.DL[1:0] bits. 16-byte boundary: Lower 4 bits = 0000b 32-byte boundary: Lower 5 bits = 00000b 64-byte boundary: Lower 6 bits = 000000b	R/W

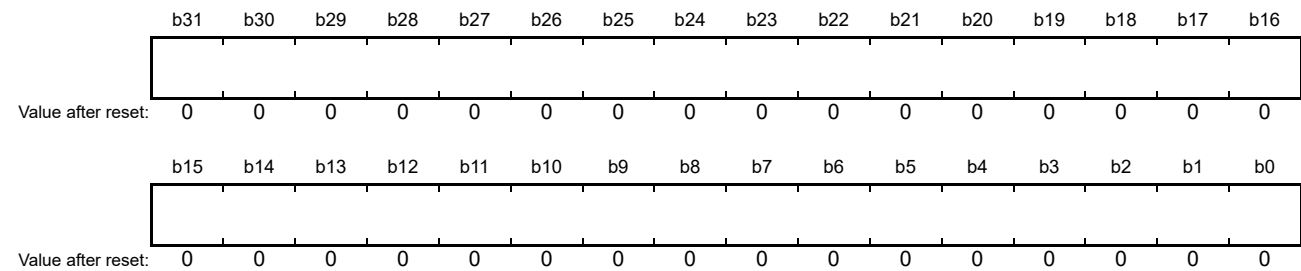
The TDLAR register sets the start address of the transmit descriptor list.

Align each descriptor on the corresponding boundary to the descriptor length selected by the EDMR.DL[1:0] bits.

Do not rewrite the TDLAR register during transmission. Rewrite the TDLAR register while the EDTRR.TR bit is 0.

38.2.5 Receive Descriptor List Start Address Register (RDLAR)

Address(es): EDMAC0.RDLAR 000C 0020h, EDMAC1.RDLAR 000C 0220h, PTPEDMAC.RDLAR 000C 0420h



Bit	Symbol	Bit Name	Description	R/W
b31 to b0	—	—	The start address of the receive descriptor list is set. Set the start address according to the descriptor length selected by the EDMR.DL[1:0] bits. 16-byte boundary: Lower 4 bits = 0000b 32-byte boundary: Lower 5 bits = 00000b 64-byte boundary: Lower 6 bits = 000000b	R/W

The RDLAR register sets the start address of the receive descriptor list.

Align each descriptor on the corresponding boundary to the descriptor length selected by the EDMR.DL[1:0] bits.

Do not rewrite the RDLAR register during reception. Rewrite the RDLAR register while the EDRRR.RR bit is 0.

38.2.6 ETHERC/EDMAC Status Register (EDMACn.EESR)

Address(es): EDMAC0.EESR 000C 0028h, EDMAC1.EESR 000C 0228h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	TWB	—	—	—	TABT	RABT	RFCOF	—	ECI	TC	TDE	TFUF	FR	RDE	RFOF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	CND	DLC	CD	TRO	RMAF	—	—	RRF	RTLF	RTSF	PRE	CERF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CERF	CRC Error Flag	0: CRC error has not been detected. 1: CRC error has been detected.	R/W
b1	PRE	PHY-LSI Receive Error Flag	0: PHY-LSI receive error has not been detected. 1: PHY-LSI receive error has been detected.	R/W
b2	RTSF	Frame-Too-Short Error Flag	0: Frame-too-short error has not been detected. 1: Frame-too-short error has been detected.	R/W
b3	RTLF	Frame-Too-Long Error Flag	0: Frame-too-long error has not been detected. 1: Frame-too-long error has been detected.	R/W
b4	RRF	Alignment Error Flag	0: Alignment error has not been detected. 1: Alignment error has been detected.	R/W
b6, b5	—	Reserved	The read value is 0. The write value should be 0.	R/W
b7	RMAF	Multicast Address Frame Receive Flag	0: Multicast address frame has not been received. 1: Multicast address frame has been received.	R/W
b8	TRO	Transmit Retry Over Flag	0: Transmit retry-over condition has not been detected. 1: Transmit retry-over condition has been detected.	R/W
b9	CD	Late Collision Detect Flag	0: Late collision has not been detected. 1: Late collision has been detected during frame transmission.	R/W
b10	DLC	Loss of Carrier Detect Flag	0: Loss of carrier has not been detected. 1: Loss of carrier has been detected during frame transmission.	R/W
b11	CND	Carrier Not Detect Flag	0: A carrier has been detected when transmission starts. 1: A carrier has not been detected during preamble transmission.	R/W
b15 to b12	—	Reserved	The read value is 0. The write value should be 0.	R/W
b16	RFOF	Receive FIFO Overflow Flag	0: Overflow has not occurred. 1: Overflow has occurred.	R/W
b17	RDE	Receive Descriptor Empty Flag	0: The EDMAC detects that the receive descriptor valid bit (RD0.RACT) is 1. 1: The EDMAC detects that the receive descriptor valid bit (RD0.RACT) is 0.	R/W
b18	FR	Frame Receive Flag	0: Frame has not been received. 1: Frame has been received. Update of the receive descriptor is complete.	R/W
b19	TFUF	Transmit FIFO Underflow Flag	0: Underflow has not occurred. 1: Underflow has occurred.	R/W
b20	TDE	Transmit Descriptor Empty Flag	0: The EDMAC detects that the transmit descriptor valid bit (TD0.TACT) is 1. 1: The EDMAC detects that the transmit descriptor valid bit (TD0.TACT) is 0.	R/W

Bit	Symbol	Bit Name	Description	R/W
b21	TC	Frame Transfer Complete Flag	0: Transfer have not been completed, or no transfer has been requested. 1: All frames indicated by the transmit descriptor have been completely transferred to the transmit FIFO.	R/W
b22	ECI	ETHERC Status Register Source Flag	0: ETHERC status interrupt source has not been detected. 1: ETHERC status interrupt source has been detected.	R *1
b23	—	Reserved	The read value is 0. The write value should be 0.	R/W
b24	RFCOF	Receive Frame Counter Overflow Flag	0: Receive frame counter has not overflowed. 1: Receive frame counter has overflowed.	R/W
b25	RABT	Receive Abort Detect Flag	0: Frame reception has not been aborted or no reception has been requested. 1: Frame reception has been aborted.	R/W
b26	TABT	Transmit Abort Detect Flag	0: Frame transmission has not been aborted or no transmission has been requested. 1: Frame transmission has been aborted.	R/W
b29 to b27	—	Reserved	The read value is 0. The write value should be 0.	R/W
b30	TWB	Write-Back Complete Flag	0: Write-back has not been completed, or no transmission has been requested. 1: Write-back to the transmit descriptor has been completed.	R/W
b31	—	Reserved	The read value is 0. The write value should be 0.	R/W

Note 1. The ECI flag is read-only. When the source in the ETHERCn.ECSR register is cleared, the ECI flag is also cleared.

The EDMACn.EESR register indicates the ETHERCn and EDMACn communication status.

Each flag in the EESR register can be output as an interrupt request signal (EINTn) from the EDMACn. Each flag, excluding the ECI flag, becomes 0 by writing 1. The value of each flag is not changed by writing 0. Each interrupt source can be enabled by the corresponding bit in the EDMACn.EESIPR register.

CERF Flag (CRC Error Flag)

The CERF flag becomes 1 when an error is detected in checking the frame check sequence (FCS) field of the receive frame.

PRE Flag (PHY-LSI Receive Error Flag)

The PRE flag indicates the RX_ER signal output from the PHY-LSI is high.

RTSF Flag (Frame-Too-Short Error Flag)

The RTSF flag indicates that a received frame was less than 64 bytes.

RTLFL Flag (Frame-Too-Long Error Flag)

The RTLFL flag indicates that a received frame is greater than the upper limit of the receive frame length set in the ETHERCn.RFLR register. The excess data is discarded.

RRF Flag (Alignment Error Flag)

The RRF flag indicates that a frame is not an integral number of octets. The last word that is not an integral number of octets is not transferred.

An error frame shorter than 16 bytes is discarded and the RRF flag does not become 1.

RMAF Flag (Multicast Address Frame Receive Flag)

The RMAF flag indicates that a multicast frame has been received.

TRO Flag (Transmit Retry Over Flag)

This flag indicates that a collision occurred again during the 15th retry of frame transmission.

CD Flag (Late Collision Detect Flag)

The CD flag indicates that a late collision has been detected during frame transmission.

DLC Flag (Loss of Carrier Detect Flag)

The DLC flag indicates that a loss of carrier has been detected during frame transmission.

CND Flag (Carrier Not Detect Flag)

The CND flag becomes 1 when a carrier has not been detected during preamble transmission.

RFOF Flag (Receive FIFO Overflow Flag)

The RFOF flag indicates that the receive FIFO has overflowed during frame reception.

RDE Flag (Receive Descriptor Empty Flag)

The RDE flag indicates that the read receive descriptor is invalid.

When this flag becomes 1, set the RD0.RACT bit in the receive descriptor to 1 and set the EDRRR.RR bit to 1 to resume reception.

FR Flag (Frame Receive Flag)

The FR flag indicates that a frame has been received and the receive descriptor has been updated. The FR flag becomes 1 every time a frame is received.

TFUF Flag (Transmit FIFO Underflow Flag)

The TFUF flag indicates that no data remains in the transmit FIFO during frame transmission. Incomplete data is sent on the line.

TDE Flag (Transmit Descriptor Empty Flag)

The TDE flag indicates that the TD0.TACT bit of the transmit descriptor is 0 while the previous transmit descriptor indicates that the frame is not complete (TD0.TFP[1:0] bits are 10b or 00b) in multi-buffer frame transmission. As a result, an incomplete frame may be sent.

When this flag becomes 1, perform a software reset and then set the EDTRR.TR bit to 1 to resume transmission.

Transmission starts from the address stored in the TDLAR register.

TC Flag (Frame Transfer Complete Flag)

The TC flag indicates that all the data specified by the transmit descriptor has been transmitted from the ETHERC. This flag becomes 1 when one frame has been transmitted in single-buffer frame transmission or when the last data of a frame is transmitted in multi-buffer frame transmission and the TD0.TACT bit in the next transmit descriptor is 0. After frame transmission is completed, the EDMAC writes the transfer status back to the descriptor.

ECI Flag (ETHERC Status Register Source Flag)

The ECI flag becomes 1 when an interrupt request is generated by the ETHERCn.ECSR register.

RFCOF Flag (Receive Frame Counter Overflow Flag)

The RFCOF flag indicates that the next frame reception starts while the number of frames stored in the receive FIFO reaches the maximum number of frames (16 frames). Note that the received frame is discarded while the RFCOF flag is 1.

RABT Flag (Receive Abort Detect Flag)

The RABT flag indicates that the ETHERC has aborted frame reception due to a CRC error, PHY-LSI receive error, frame-too-short error, frame-too-long error, or other error.

An error frame shorter than 16 bytes is discarded and the RABT flag does not become 1.

TABT Flag (Transmit Abort Detect Flag)

The TABT flag indicates that the ETHERC has aborted frame transmission due to transmit retry over, loss of carrier, no carrier detection, or other error.

TWB Flag (Write-Back Complete Flag)

The TWB flag indicates the EDMAC has completed writing back to the descriptor after frame transmission. Note that this flag becomes 1 after each frame transmission when the TRIMD.TIM bit is 0. This flag becomes 1 only when the TRIMD.TIS bit is 1.

38.2.7 PTP/EDMAC Status Register (PTPEDMAC.EESR)

Address(es): PTPEDMAC.EESR 000C 0428h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	TWB	—	—	—	TABT	—	RFCOF	—	—	TC	TDE	TFUF	FR	RDE	RFOF
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	MACE	RPORT	—	—	PVER	TYPE[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	TYPE[3:0]	PTP v2 Message Type Flag	b3 b0 0 0 0 0: Sync 0 0 0 1: Delay_Req 0 0 1 0: Pdelay_Req 0 0 1 1: Pdelay_Resp 1 0 0 0: Follow_Up 1 0 0 1: Delay_Resp 1 0 1 0: Pdelay_Resp_Follow_Up 1 0 1 1: Announce 1 1 0 0: Signaling 1 1 0 1: Management Settings other than above are reserved.	R/W
b4	PVER	PTP v2 Packet Flag	0: The current packet is not a PTP v2 packet. 1: The current packet is a PTP v2 packet.	R/W
b6, b5	—	Reserved	The read value is 0. The write value should be 0.	R/W
b7	RPORT	Receive Port Flag	0: Port 0 1: Port 1	R/W
8	MACE	MAC Address Mismatch Flag	0: The source MAC address of transmit frame data matches the set value. 1: The source MAC address of transmit frame data does not match the set value.	R/W
b15 to b9	—	Reserved	The read value is 0. The write value should be 0.	R/W
b16	RFOF	Receive FIFO Overflow Flag	0: Overflow has not occurred. 1: Overflow has occurred.	R/W
b17	RDE	Receive Descriptor Empty Flag	0: The EDMAC detects that the receive descriptor valid bit (RD0.RACT) is 1. 1: The EDMAC detects that the receive descriptor valid bit (RD0.RACT) is 0.	R/W
b18	FR	Frame Receive Flag	0: Frame has not been received. 1: Frame has been received. The receive descriptor has been updated.	R/W
b19	TFUF	Transmit FIFO Underflow Flag	0: Underflow has not occurred. 1: Underflow has occurred.	R/W
b20	TDE	Transmit Descriptor Empty Flag	0: The EDMAC detects that the transmit descriptor valid bit (TD0.TACT) is 1. 1: The EDMAC detects that the transmit descriptor valid bit (TD0.TACT) is 0.	R/W
b21	TC	Frame Transfer Complete Flag	0: Transfer has not been completed, or transfer has not been requested. 1: All frames indicated by the transmit descriptor have been completely transferred to the transmit FIFO.	R/W
b23, b22	—	Reserved	The read value is 0. The write value should be 0.	R/W
b24	RFCOF	Receive Frame Counter Overflow Flag	0: Receive frame counter has not overflowed. 1: Receive frame counter has overflowed.	R/W

Bit	Symbol	Bit Name	Description	R/W
b25	—	Reserved	The read value is 0. The write value should be 0.	R/W
b26	TABT	Transmit Abort Detect Flag	0: Frame transmission has not been aborted or transmission has not been requested. 1: Frame transmission has been aborted.	R/W
b29 to b27	—	Reserved	The read value is 0. The write value should be 0.	R/W
b30	TWB	Write-Back Complete Flag	0: Write-back has not been completed, or transmission has not been requested. 1: Write-back to the transmit descriptor has been completed.	R/W
b31	—	Reserved	The read value is 0. The write value should be 0.	R/W

The PTPEDMAC.EESR register indicates the PTPEDMAC communication status.

Each flag in the EESR register can be output as an interrupt request signal (PINT) from the PTPEDMAC. Each flag becomes 0 by writing 1. The value of each flag is not changed by writing 0. Each interrupt source, excluding the TYPE[3:0] flag, can be enabled by setting the corresponding bit in the PTPEDMAC.EESIPR register.

TYPE[3:0] Flags (PTP v2 Message Type Flag)

The TYPE[3:0] flags indicate the type of received PTP message.

PVER Flag (PTP v2 Packet Flag)

The PVER flag indicates whether the received packet is a PTP v2 packet.

RPORT Flag (Receive Port Flag)

The RPORT flag indicates which Ethernet port has been used for receiving PTP messages.

MACE Flag (MAC Address Mismatch Flag)

The MACE flag indicates the source MAC address is different from the set value.

RFOF Flag (Receive FIFO Overflow Flag)

The RFOF flag indicates that the receive FIFO has overflowed during frame reception.

RDE Flag (Receive Descriptor Empty Flag)

The RDE flag indicates that the read receive descriptor is invalid.

When this flag becomes 1, set the RD0.RACT bit in the receive descriptor to 1 and set the EDTRR.RR bit to 1 to resume reception.

FR Flag (Frame Receive Flag)

The FR flag indicates that a frame has been received and the receive descriptor has been updated. The FR flag becomes 1 every time a frame is received.

TFUF Flag (Transmit FIFO Underflow Flag)

The TFUF flag indicates that no data remains in the transmit FIFO during frame transmission. Incomplete data is sent on the line.

TDE Flag (Transmit Descriptor Empty Flag)

The TDE flag indicates that the TD0.TACT bit of the transmit descriptor is 0 while the previous transmit descriptor indicates that the frame is not complete (TD0.TFP[1:0] bits are 10b or 00b) in multi-buffer frame transmission. As a result, an incomplete frame may be sent.

When this flag becomes 1, perform a software reset and then set the EDTRR.TR bit to 1 to resume transmission.

Transmission starts from the address stored in the TDLAR register.

TC Flag (Frame Transfer Complete Flag)

The TC flag indicates that all the data specified by the transmit descriptor has been transmitted from the ETHERC. This flag becomes 1 when one frame has been transmitted in single-buffer frame transmission or when the last data of a frame is transmitted in multi-buffer frame transmission and the TD0.TACT bit in the next transmit descriptor is 0. After frame transmission is completed, the PTPEDMAC writes the transfer status back to the descriptor.

RFCOF Flag (Receive Frame Counter Overflow Flag)

The RFCOF flag indicates that the next frame reception starts while the number of frames stored in the receive FIFO reaches the maximum number of frames (16 frames). Note that the received frame is discarded while the RFCOF flag is 1.

TABT Flag (Transmit Abort Detect Flag)

The TABT flag indicates that the ETHERC has aborted frame transmission due to transmit retry over, loss of carrier, no carrier detection, or other error.

TWB Flag (Write-Back Complete Flag)

The TWB flag indicates the PTPEDMAC has completed writing back to the descriptor after frame transmission. Note that this flag becomes 1 after each frame transmission when the TRIMD.TIM bit is 0. This flag becomes 1 only when the TRIMD.TIS bit is 1.

38.2.8 ETHERC/EDMAC Status Interrupt Enable Register (EDMACn.EESIPR)

Address(es): EDMAC0.EESIPR 000C 0030h, EDMAC1.EESIPR 000C 0230h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	TWBIP	—	—	—	TABTIP	RABTIP	RFCOFIP	—	ECIIP	TCIP	TDEIP	TFUFIP	FRIP	RDEIP	RFOFIP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	CNDIP	DLCIP	CDIP	TROIP	RMAFIP	—	—	RRFIP	RTLFIP	RTSFIP	PREIP	CERFIP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	CERFIP	CRC Error Interrupt Request Enable	0: CRC error interrupt request is disabled. 1: CRC error interrupt request is enabled.	R/W
b1	PREIP	PHY-LSI Receive Error Interrupt Request Enable	0: PHY-LSI receive error interrupt request is disabled. 1: PHY-LSI receive error interrupt request is enabled.	R/W
b2	RTSFIP	Frame-Too-Short Error Interrupt Request Enable	0: Frame-too-short error interrupt request is disabled. 1: Frame-too-short error interrupt request is enabled.	R/W
b3	RTLFIP	Frame-Too-Long Error Interrupt Request Enable	0: Frame-too-long error interrupt request is disabled. 1: Frame-too-long error interrupt request is enabled.	R/W
b4	RRFIP	Alignment Error Interrupt Request Enable	0: Alignment error interrupt request is disabled. 1: Alignment error interrupt request is enabled.	R/W
b6, b5	—	Reserved	The read value is 0. The write value should be 0.	R/W
b7	RMAFIP	Multicast Address Frame Receive Interrupt Request Enable	0: Multicast address frame receive interrupt request is disabled. 1: Multicast address frame receive interrupt request is enabled.	R/W
b8	TROIP	Transmit Retry Over Interrupt Request Enable	0: Transmit retry over interrupt request is disabled. 1: Transmit retry over interrupt request is enabled.	R/W
b9	CDIP	Late Collision Detect Interrupt Request Enable	0: Late collision detect interrupt request is disabled. 1: Late collision detect interrupt request is enabled.	R/W
b10	DLCIP	Loss of Carrier Detect Interrupt Request Enable	0: Loss of carrier detect interrupt request is disabled. 1: Loss of carrier detect interrupt request is enabled.	R/W
b11	CNDIP	Carrier Not Detect Interrupt Request Enable	0: Carrier not detect interrupt request is disabled. 1: Carrier not detect interrupt request is enabled.	R/W
b15 to b12	—	Reserved	The read value is 0. The write value should be 0.	R/W
b16	RFOFIP	Receive FIFO Overflow Interrupt Request Enable	0: Overflow interrupt request is disabled. 1: Overflow interrupt request is enabled.	R/W
b17	RDEIP	Receive Descriptor Empty Interrupt Request Enable	0: Receive descriptor empty interrupt request is disabled. 1: Receive descriptor empty interrupt request is enabled.	R/W
b18	FRIP	Frame Receive Interrupt Request Enable	0: Frame reception interrupt request is disabled. 1: Frame reception interrupt request is enabled.	R/W
b19	TFUFIP	Transmit FIFO Underflow Interrupt Request Enable	0: Underflow interrupt request is disabled. 1: Underflow interrupt request is enabled.	R/W
b20	TDEIP	Transmit Descriptor Empty Interrupt Request Enable	0: Transmit descriptor empty interrupt request is disabled. 1: Transmit descriptor empty interrupt request is enabled.	R/W
b21	TCIP	Frame Transfer Complete Interrupt Request Enable	0: Frame transmission complete interrupt request is disabled. 1: Frame transmission complete interrupt request is enabled.	R/W

Bit	Symbol	Bit Name	Description	R/W
b22	ECIIP	ETHERC Status Register Source Interrupt Request Enable	0: ETHERC status interrupt request is disabled. 1: ETHERC status interrupt request is enabled.	R/W
b23	—	Reserved	The read value is 0. The write value should be 0.	R/W
b24	RFCOFIP	Receive Frame Counter Overflow Interrupt Request Enable	0: Receive frame counter overflow interrupt request is disabled. 1: Receive frame counter overflow interrupt request is enabled.	R/W
b25	RABTIP	Receive Abort Detect Interrupt Request Enable	0: Receive abort detect interrupt request is disabled. 1: Receive abort detect interrupt request is enabled.	R/W
b26	TABTIP	Transmit Abort Detect Interrupt Request Enable	0: Transmit abort detect interrupt request is disabled. 1: Transmit abort detect interrupt request is enabled.	R/W
b29 to b27	—	Reserved	The read value is 0. The write value should be 0.	R/W
b30	TWBIP	Write-Back Complete Interrupt Request Enable	0: Write-back complete interrupt request is disabled. 1: Write-back complete interrupt request is enabled.	R/W
b31	—	Reserved	The read value is 0. The write value should be 0.	R/W

The EDMACn.EESIPR register enables interrupt requests corresponding to bits in the EDMACn.EESR register. When a bit in this register is 1, the corresponding interrupt request is enabled.

38.2.9 PTP/EDMAC Status Interrupt Enable Register (PTPEDMAC.EESIPR)

Address(es): PTPEDMAC.EESIPR 000C 0430h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	TWBIP	—	—	—	TABTIP	—	RFCOFIP	—	—	TCIP	TDEIP	TFUFIP	FRIP	RDEIP	RFOFIP
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	MACEIP	RPORTIP	—	—	PVERIP	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	—	Reserved	The read value is 0. The write value should be 0.	R/W
b4	PVERIP	PTP v2 Packet Receive Interrupt Request Enable	0: PTP v2 packet receive interrupt request is disabled. 1: PTP v2 packet receive interrupt request is enabled.	R/W
b6, b5	—	Reserved	The read value is 0. The write value should be 0.	R/W
b7	RPORTIP	Receive Port Interrupt Request Enable	0: Interrupt request at a fame reception on port 1 is disabled. 1: Interrupt request at a fame reception on port 1 is enabled.	R/W
b8	MACEIP	MAC Address Mismatch Interrupt Request Enable	0: This bit disables an interrupt request generated when the source MAC address of transmit frame data does not match the set value. 1: This bit enables an interrupt request generated when the source MAC address of transmit frame data does not match the set value.	R/W
b15 to b9	—	Reserved	The read value is 0. The write value should be 0.	R/W
b16	RFOFIP	Receive FIFO Overflow Interrupt Request Enable	0: Overflow interrupt request is disabled. 1: Overflow interrupt request is enabled.	R/W
b17	RDEIP	Receive Descriptor Empty Interrupt Request Enable	0: Receive descriptor empty interrupt request is disabled. 1: Receive descriptor empty interrupt request is enabled.	R/W
b18	FRIP	Frame Receive Interrupt Request Enable	0: Frame receive interrupt request is disabled. 1: Frame receive interrupt request is enabled.	R/W
b19	TFUFIP	Transmit FIFO Underflow Interrupt Request Enable	0: Underflow interrupt request is disabled. 1: Underflow interrupt request is enabled.	R/W
b20	TDEIP	Transmit Descriptor Empty Interrupt Request Enable	0: Transmit descriptor empty interrupt request is disabled. 1: Transmit descriptor empty interrupt request is enabled.	R/W
b21	TCIP	Frame Transfer Complete Interrupt Request Enable	0: Frame transmission complete interrupt request is disabled. 1: Frame transmission complete interrupt request is enabled.	R/W
b23, b22	—	Reserved	The read value is 0. The write value should be 0.	R/W
b24	RFCOFIP	Receive Frame Counter Overflow Interrupt Request Enable	0: Receive frame counter overflow interrupt request is disabled. 1: Receive frame counter overflow interrupt request is enabled.	R/W
b25	—	Reserved	The read value is 0. The write value should be 0.	R/W
b26	TABTIP	Transmit Abort Detect Interrupt Request Enable	0: Transmit abort detect interrupt request is disabled. 1: Transmit abort detect interrupt request is enabled.	R/W
b29 to b27	—	Reserved	The read value is 0. The write value should be 0.	R/W

Bit	Symbol	Bit Name	Description	R/W
b30	TWBIP	Write-Back Complete Interrupt Request Enable	0: Write-back complete interrupt request is disabled. 1: Write-back complete interrupt request is enabled.	R/W
b31	—	Reserved	The read value is 0. The write value should be 0.	R/W

The PTPEDMAC.EESIPR register enables interrupt requests corresponding to bits in the PTPEDMAC.EESR register. When a bit in this register is 1, the corresponding interrupt request is enabled.

38.2.10 ETHERC/EDMAC Transmit/Receive Status Copy Enable Register (EDMACn.TRSCER)

Address(es): EDMAC0.TRSCER 000C 0038h, EDMAC1.TRSCER 000C 0238h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

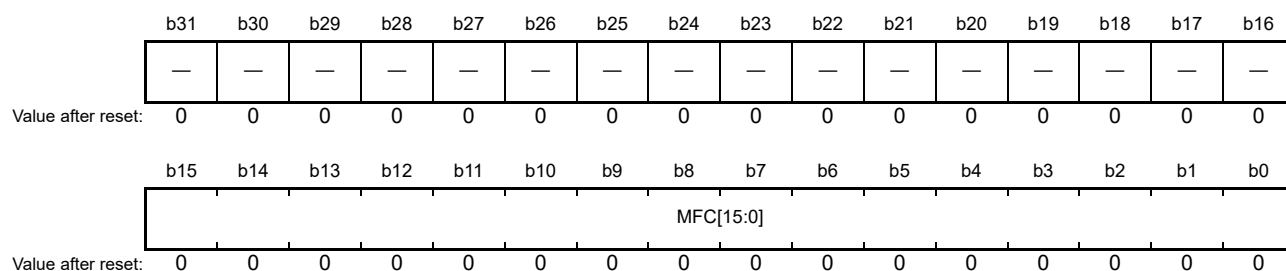
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	RMAFCE	—	—	RRFCE	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	—	Reserved	The read value is 0. The write value should be 0.	R/W
b4	RRFCE	RRF Flag Copy Enable	0: The EESR.RRF flag status is reflected in the RD0.RFE bit of the receive descriptor. 1: The EESR.RRF flag status is not reflected in the RD0.RFE bit of the receive descriptor.	R/W
b6, b5	—	Reserved	The read value is 0. The write value should be 0.	R/W
b7	RMAFCE	RMAF Flag Copy Enable	0: The EESR.RMAF flag status is reflected in the RD0.RFE bit of the receive descriptor. 1: The EESR.RMAF flag status is not reflected in the RD0.RFE bit of the receive descriptor.	R/W
b31 to b8	—	Reserved	The read value is 0. The write value should be 0.	R/W

The EDMACn.TRSCER register selects whether the receive status indicated by flags EDMACn.EESR.RMAF and RRF is reflected in the RFE bit of the receive descriptor as a summary. The bits in the TRSCER register correspond to bits in the EESR register that have the same number. When setting the RMAFCE or RRFCE bit to 0, the corresponding receive status (EESR.RMAF or RRF flag) is reflected in the RFE bit of the receive descriptor. When setting the RMAFCE or RRFCE bit to 1, the corresponding receive status is not reflected in the RFE bit.

38.2.11 Missed-Frame Counter Register (RMFCR)

Address(es): EDMAC0.RMFCR 000C 0040h, EDMAC1.RMFCR 000C 0240h, PTPEDMAC.RMFCR 000C 0440h



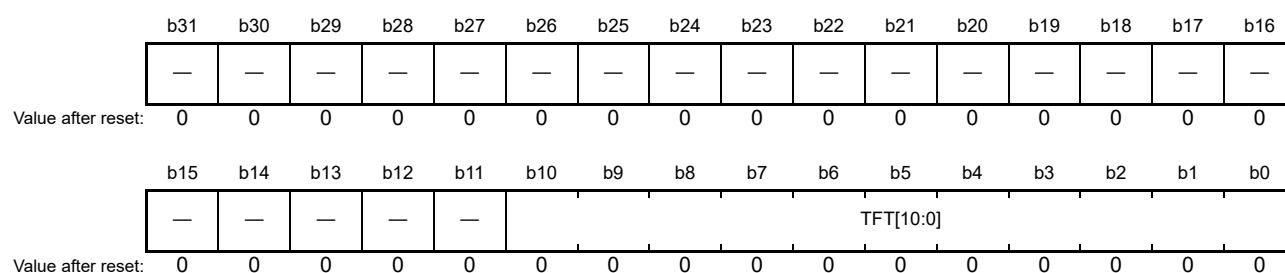
Bit	Symbol	Bit Name	Description	R/W
b15 to b0	MFC[15:0]	Missed-Frame Counter	These bits indicate the number of frames that are discarded and not transferred to the receive buffer during reception.	R/W
b31 to b16	—	Reserved	The read value is 0. The write value should be 0.	R/W

The RMFCR register indicates that the number of frames that could not be stored in the receive FIFO and were therefore discarded during reception. When the receive FIFO overflows, it stops receiving data, and the remaining frames are discarded. At the same time, the RMFCR register value is incremented. When the RMFCR register value reaches FFFFh, incrementing is halted. When any value is written to the RMFCR register, the counter value becomes 0.

For the frame that has not been completely received, after data in the receive FIFO is transferred to the receive buffer, the RACT bit in the receive descriptor 0 (RD0) becomes 0 (descriptor disabled), the RFS9 bit becomes 1 (receive FIFO overflow), and the EDMACn.EESR.RFOF or PTPEDMAC.EESR.RFOF flag becomes 1 (overflow has occurred).

38.2.12 Transmit FIFO Threshold Register (TFTR)

Address(es): EDMAC0.TFTR 000C 0048h, EDMAC1.TFTR 000C 0248h, PTPEDMAC.TFTR 000C 0448h



Bit	Symbol	Bit Name	Description	R/W
b10 to b0	TFT[10:0]	Transmit FIFO Threshold	000h: Store and forward mode 001h to 00Ch: Setting prohibited 00Dh to 200h: The threshold is the set value multiplied by 4. Example: 00Dh: 52 bytes 040h: 256 bytes 100h: 1024 bytes 200h: 2048 bytes 201h to 7FFh: Setting prohibited	R/W
b31 to b11	—	Reserved	The read value is 0. The write value should be 0.	R/W

Note 1. When starting transmission before one frame data has been completely written, take care to prevent an underflow.

Note 2. To prevent a transmit underflow, using the initial value (store and forward mode) is recommended.

The TFTR register sets the transmit FIFO threshold at which the first transmission starts. The actual threshold is the set value multiplied by 4.

The ETHERC starts transmission when the amount of data in the transmit FIFO exceeds the number of bytes set in the TFTR register, when the transmit FIFO is full, or when one frame data has been completely written. Set the TFTR register while the EDTRR.TR bit is 0.

38.2.13 FIFO Depth Register (FDR)

Address(es): EDMAC0.FDR 000C 0050h, EDMAC1.FDR 000C 0250h, PTPEDMAC.FDR 000C 0450h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—			TFD[4:0]			—	—	—			RFD[4:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b4 to b0	RFD[4:0]	Receive FIFO Depth	b4 b0 01111: 4096 bytes Settings other than above are prohibited.	R/W
b7 to b5	—	Reserved	The read value is 0. The write value should be 0.	R/W
b12 to b8	TFD[4:0]	Transmit FIFO Depth	b12 b8 00111: 2048 bytes Settings other than above are prohibited.	R/W
b31 to b13	—	Reserved	The read value is 0. The write value should be 0.	R/W

The FDR register sets the transmit and receive FIFO depths.

Set this register to 0000 070Fh before starting transmission and reception.

38.2.14 Receive Method Control Register (RMCR)

Address(es): EDMAC0.RMCR 000C 0058h, EDMAC1.RMCR 000C 0258h, PTPEDMAC.RMCR 000C 0458h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RNR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	RNR	Receive Request Reset	0: EDRRR.RR bit (receive request bit) is set to 0 when one frame has been received. 1: EDRRR.RR bit (receive request bit) is not set to 0 when one frame has been received.	R/W
b31 to b1	—	Reserved	The read value is 0. The write value should be 0.	R/W

The RMCR register sets how to control the EDRRR.RR bit when receiving a frame.

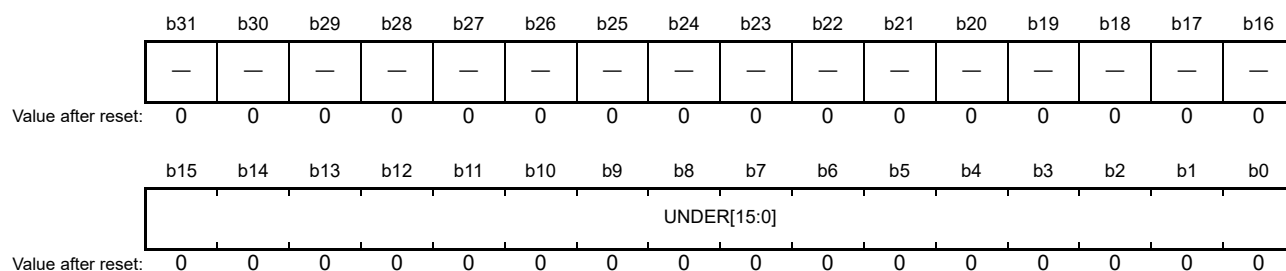
Since the EDRRR.RR bit becomes 0 when one frame has been received while the RNR bit is 0, the RR bit needs to be set to 1 by software to receive the subsequent frame.

Since the EDRRR.RR bit does not become 0 when one frame has been received while the RNR bit is 1, the EDMAC reads the next receive descriptor and continues receiving frames. However, the EDRRR.RR bit becomes 0 when the RD0.RACT bit in the receive descriptor is 0. It is recommended to set the RNR bit to 1 when receiving data continuously.

Set the RMCR register while the EDRRR.RR bit is 0.

38.2.15 Transmit FIFO Underflow Counter (TFUCR)

Address(es): EDMAC0.TFUCR 000C 0064h, EDMAC1.TFUCR 000C 0264h, PTPEDMAC.TFUCR 000C 0464h



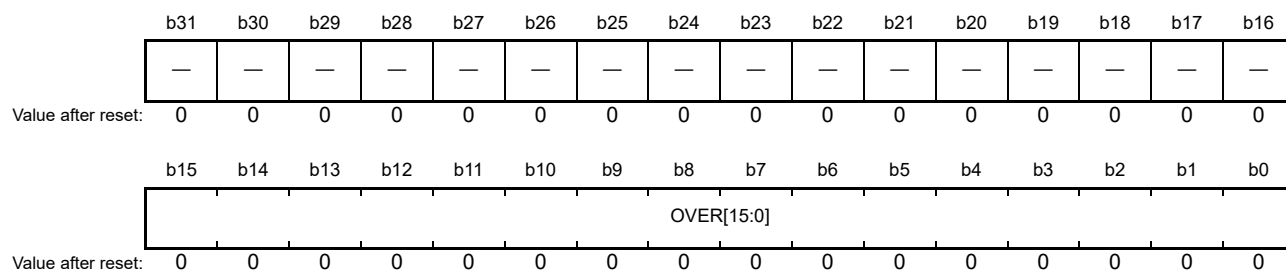
Bit	Symbol	Bit Name	Description	R/W
b15 to b0	UNDER[15:0]	Transmit FIFO Underflow Count	These bits indicate how many times the transmit FIFO has underflowed. The counter stops when the counter value reaches FFFFh.	R/W
b31 to b16	—	Reserved	The read value is 0. The write value should be 0.	R/W

The TFUCR register indicates how many times the transmit FIFO has underflowed.

When writing any value to the TFUCR register, the counter value becomes 0.

38.2.16 Receive FIFO Overflow Counter (RFOCR)

Address(es): EDMAC0.RFOCR 000C 0068h, EDMAC1.RFOCR 000C 0268h, PTPEDMAC.RFOCR 000C 0468h



Bit	Symbol	Bit Name	Description	R/W
b15 to b0	OVER[15:0]	Receive FIFO Overflow Count	These bits indicate how many times the receive FIFO has overflowed. The counter stops when the counter value reaches FFFFh.	R/W
b31 to b16	—	Reserved	The read value is 0. The write value should be 0.	R/W

The RFOCR register indicates how many times the receive FIFO has overflowed.

When writing any value to the RFOCR register, the counter value becomes 0.

38.2.17 Independent Output Signal Setting Register (IOSR)

Address(es): EDMAC0.IOSR 000C 006Ch, EDMAC1.IOSR 000C 026Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ELB
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	ELB	External Loopback Mode	0: The ETn_EXOUT pin outputs low. 1: The ETn_EXOUT pin outputs high.	R/W
b31 to b1	—	Reserved	The read value is 0. The write value should be 0.	R/W

The IOSR register selects the output level of the ETHERC external output pin (ETn_EXOUT) in external loopback mode.

The ELB bit value is output to the ETn_EXOUT pin of the MCU as is, which can be used to set loopback mode for the PHY-LSI. To use the loopback function of the PHY-LSI through this register, the PHY-LSI needs to be provided with the pin to be connected to the ETn_EXOUT pin.

38.2.18 Flow Control Start FIFO Threshold Setting Register (FCFTR)

Address(es): EDMAC0.FCFTR 000C 0070h, EDMAC1.FCFTR 000C 0270h, PTPEDMAC.FCFTR 000C 0470h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	RFFO[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	RFDO[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1

Bit	Symbol	Bit Name	Description	R/W
b2 to b0	RFDO[2:0]	Receive FIFO Data PAUSE Output Threshold	b2 b0 0 0 0: When 224 (256 - 32) bytes of data is stored in the receive FIFO. 0 0 1: When 480 (512 - 32) bytes of data is stored in the receive FIFO. : 1 1 0: When 1760 (1792 - 32) bytes of data is stored in the receive FIFO. 1 1 1: When 2016 (2048 - 32) bytes of data is stored in the receive FIFO.	R/W
b15 to b3	—	Reserved	The read value is 0. The write value should be 0.	R/W
b18 to b16	RFFO[2:0]	Receive FIFO Frame PAUSE Output Threshold	b18 b16 0 0 0: When 2 receive frames have been stored in the receive FIFO. 0 0 1: When 4 receive frames have been stored in the receive FIFO. 0 1 0: When 6 receive frames have been stored in the receive FIFO. : 1 1 0: When 14 receive frames have been stored in the receive FIFO. 1 1 1: When 16 receive frames have been stored in the receive FIFO.	R/W
b31 to b19	—	Reserved	The read value is 0. The write value should be 0.	R/W

The FCFTR register sets the ETHERC flow control (sets the threshold for automatically transmitting a PAUSE frame). The threshold can be set using the data size (RFDO[2:0] bits) and the number of frames (RFFO[2:0] bits) stored in the receive FIFO. The flow control starts when the stored data size or number of stored frames reaches its threshold.

38.2.19 Receive Data Padding Insert Register (RPADIR)

Address(es): EDMAC0.RPADIR 000C 0078h, EDMAC1.RPADIR 000C 0278h, PTPEDMAC.RPADIR 000C 0478h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PADS[1:0]
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b5 to b0	PADR[5:0]	Padding Slot	00h: Padding is inserted at the head of received data. 01h: Padding is inserted between the 1st byte and 2nd byte of received data. : 3Eh: Padding is inserted between the 62nd byte and 63rd byte of received data. 3Fh: Padding is inserted between the 63rd byte and 64th byte of received data	R/W
b15 to b6	—	Reserved	The read value is 0. The write value should be 0.	R/W
b17, b16	PADS[1:0]	Padding Size	b17 b16 0 0: No padding is inserted. 0 1: 1 byte is inserted. 1 0: 2 bytes are inserted. 1 1: 3 bytes are inserted.	R/W
b31 to b18	—	Reserved	The read value is 0. The write value should be 0.	R/W

The RPADIR register sets insertion of padding for received data. The padding value is 00h.
Set the EDMR.SWR bit to 1 to reset before rewriting the RPADIR register.

38.2.20 Transmit Interrupt Setting Register (TRIMD)

Address(es): EDMAC0.TRIMD 000C 007Ch, EDMAC1.TRIMD 000C 027Ch, PTPEDMAC.TRIMD 000C 047Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	TIM	—	—	—	TIS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	TIS	Transmit Interrupt Enable	0: Transmit Interrupt is disabled. 1: Transmit Interrupt is enabled. Set the EESR.TWB flag to 1 in the mode selected by the TIM bit to notify an interrupt.	R/W
b3 to b1	—	Reserved	The read value is 0. The write value should be 0.	R/W
b4	TIM	Transmit Interrupt Mode	0: Transmission complete interrupt mode An interrupt occurs when a frame has been transmitted. 1: Write-back complete interrupt mode An interrupt occurs when write-back to the transmit descriptor has been completed while the TWBI bit is 1.	R/W
b31 to b5	—	Reserved	The read value is 0. The write value should be 0.	R/W

The TRIMD register sets transmit interrupt mode and enables/disables the transmit interrupt.

When the condition set in this register is satisfied, the EESR.TWB flag becomes 1, and an interrupt request is output when the EESIPR.TWBIP bit is 1.

38.2.21 Receive Buffer Write Address Register (RBWAR)

Address(es): EDMAC0.RBWAR 000C 00C8h, EDMAC1.RBWAR 000C 02C8h, PTPEDMAC.RBWAR 000C 04C8h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

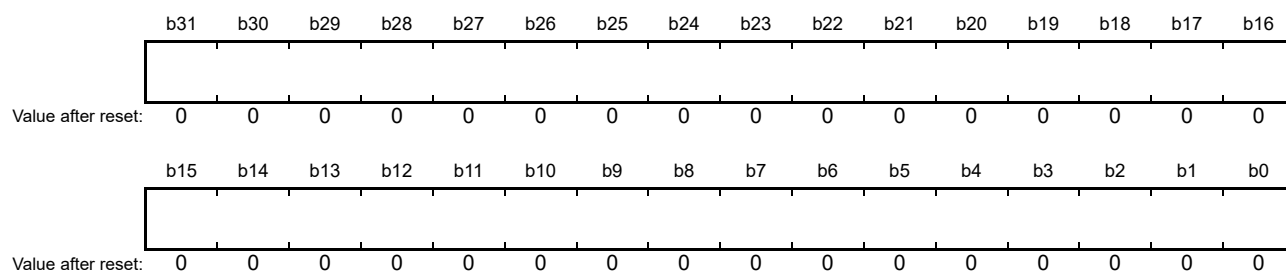
	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The RBWAR register indicates the last address that the EDMAC has written data to when writing to the receive buffer. Refer to the address indicated by the RBWAR register to recognize which address in the receive buffer the EDMAC is writing data to. Note that the address that the EDMAC is outputting to the receive buffer may not match the read value of the RBWAR register during data reception.

The RBWAR register is read only. Do not write to this register.

38.2.22 Receive Descriptor Fetch Address Register (RDFAR)

Address(es): EDMAC0.RDFAR 000C 00CCh, EDMAC1.RDFAR 000C 02CCh, PTPEDMAC.RDFAR 000C 04CCh



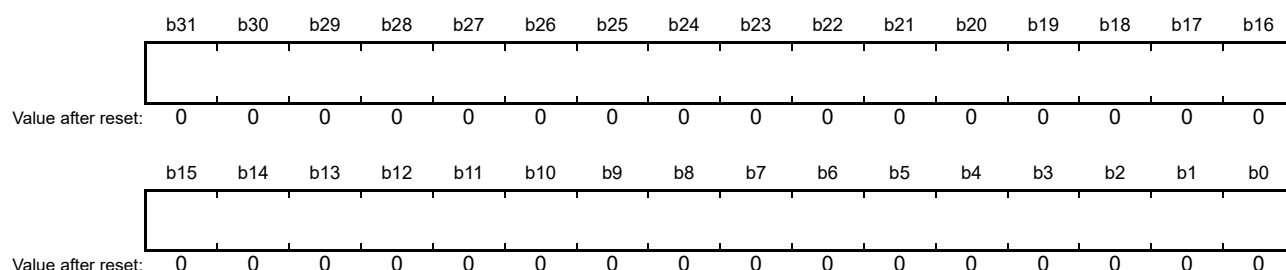
The RDFAR register indicates the start address of the last fetched receive descriptor when the EDMAC fetches descriptor information from the receive descriptor.

Refer to the address indicated by the RDFAR register to recognize which receive descriptor information the EDMAC is using for the current processing. Note that the address of the receive descriptor that the EDMAC fetches may not match the read value of the RDFAR register during data reception.

The RDFAR is read only. Do not write to this register.

38.2.23 Transmit Buffer Read Address Register (TBRAR)

Address(es): EDMAC0.TBRAR 000C 00D4h, EDMAC1.TBRAR 000C 02D4h, PTPEDMAC.TBRAR 000C 04D4h



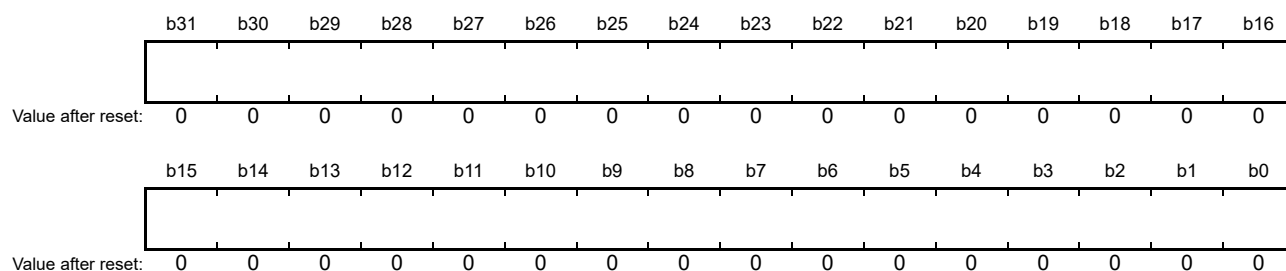
The TBRAR register indicates the last address that the EDMAC has read data from when reading data from the transmit buffer.

Refer to the address indicated by the TBRAR register to recognize which address in the transmit buffer the EDMAC is reading from. Note that the address that the EDMAC is outputting to the transmit buffer may not match the read value of the TBRAR register.

The TBRAR register is read only. Do not write to this register.

38.2.24 Transmit Descriptor Fetch Address Register (TDFAR)

Address(es): EDMAC0.TDFAR 000C 00D8h, EDMAC1.TDFAR 000C 02D8h, PTPEDMAC.TDFAR 000C 04D8h



The TDFAR register indicates the start address of the last fetched transmit descriptor when the EDMAC fetches descriptor information from the transmit descriptor.

Refer to the address indicated by the TDFAR register to recognize which transmit descriptor information the EDMAC is using for the current processing. Note that the address of the transmit descriptor that the EDMAC fetches may not match the read value of the TDFAR register.

The TDFAR is read only. Do not write to this register.

38.3 Operation

The EDMAC transfers data according to the information written in the descriptor. Two types of descriptors are provided: transmit descriptor and receive descriptor. A descriptor includes the buffer size, address, and transmit/receive status. The EDMAC transmits or receives data continuously using sequentially arranged descriptors (descriptor list).

38.3.1 Descriptor Lists and Data Buffers

To transfer data using the EDMAC, create the transmit and receive descriptor lists on the memory, set the start address of the transmit descriptor list to the TDLAR register, and set the start address of the receive descriptor list to the RDLAR register. Also, transmit and receive buffers corresponding to the each descriptor are required.

Align the descriptor list on the appropriate address boundary according to the descriptor length set by the EDMR.DL[1:0] bits. The transmit buffer can be aligned on a long word boundary, word boundary, or byte boundary. When the valid transmit buffer size is 16 bytes or less, align it on a 32-byte boundary. Align the receive buffer on a 32-byte boundary. Set different addresses for the transmit and receive descriptors and buffers for EDMAC0, EDMAC1, and the PTPEDMAC.

38.3.1.1 Transmit Descriptor

Figure 38.3 shows the relation between a transmit descriptor and transmit buffer. A transmit descriptor consists of TD0 to TD2. The transmit frame and transmit buffer configuration can be selected from one buffer per frame (single-buffer frame transmission) or multiple buffers per frame (multi-buffer frame transmission) by setting the transmit descriptor.

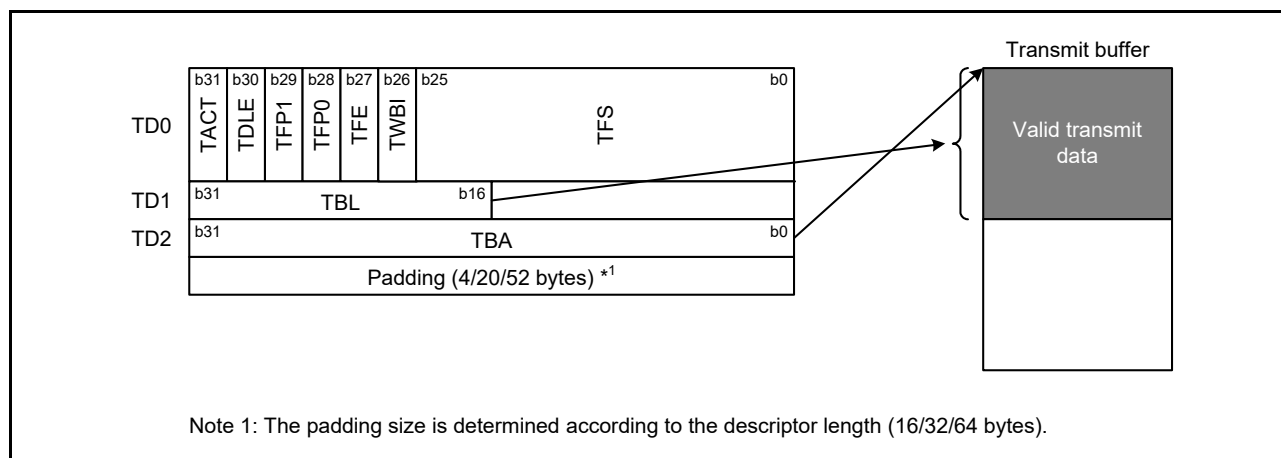


Figure 38.3 Relation between a Transmit Descriptor and Transmit Buffer

(1) Transmit Descriptor 0 (TD0)

Bit	Symbol	Bit Name	Description	R/W
<u>b25 to b0</u>	<u>TFS</u>	Transmit Frame Status	Set all bits to 0 when creating a descriptor. After write-back, the bits indicate the following: [for EDMACn] TFS25 to TFS9: Reserved TFS8: Transmit abort is detected (the value is equivalent to the EESR.TABT flag). TFS7 to TFS4: Reserved TFS3: No carrier is detected (the value is equivalent to the EESR.CND flag). TFS2: Loss of carrier is detected (the value is equivalent to the EESR.DLC flag). TFS1: Late collision during transmission is detected (the value is equivalent to the EESR.CD flag). TFS0: Transmit retry over (the value is equivalent to the EESR.TRO flag) When each bit becomes 1, it indicates that the corresponding error has occurred during frame transmission. When any of the TFS bits becomes 1, the TFE bit also becomes 1. When any bit from TFS3 to TFS0 becomes 1, the TFS8 bit also becomes 1. [for the PTPEDMAC] TFS25 to TFS9: Reserved TFS8: Transmit abort is detected (the value is equivalent to the EESR.TABT flag). TFS7 to TFS1: Reserved TFS0: The transmission source MAC address of the transmit frame data does not match the set value (the value is equivalent to the EESR.MACE flag). When each bit becomes 1, it indicates that the corresponding error has occurred during frame transmission. When any of the TFS bits becomes 1, the TFE bit also becomes 1. When TFS0 becomes 1, TFS8 also becomes 1.	R/W
b26	TWBI	Write-Back Complete Interrupt Enable	0: Interrupt does not occur when write-back to this descriptor has been completed. 1: Interrupt occurs when write-back to this descriptor has been completed.	R/W
<u>b27</u>	<u>TFE</u>	Transmit Frame Error	0: Frame transmission is successfully completed. 1: An error occurs during frame transmission (transmission aborted).	R/W
b29, b28	TFP[1:0]	Transmit Frame Position	b29 b28 0 0: Transmit buffer indicated by this descriptor is the middle of a transmit frame (frame information is incomplete). 0 1: Transmit buffer indicated by this descriptor is the end of a transmit frame (frame information is complete). 1 0: Transmit buffer indicated by this descriptor is the head of a transmit frame (frame information is incomplete.) 1 1: Transmit buffer indicated by this descriptor is all of a transmit frame (one buffer per frame).	R/W
b30	TDLE	Transmit Descriptor List End	When this bit is 1, it indicates that this descriptor is the last descriptor of the descriptor list.	R/W
<u>b31</u>	<u>TACT</u>	Transmit Descriptor Valid	This bit indicates that this descriptor is valid.	R/W

Note: Bits for write-back are underlined.

TD0 indicates the transmit frame settings, and the status after transmission.

TWBI Bit (Write-Back Complete Interrupt Enable)

This bit value is valid when the TRIMD.TIM bit is 1. Set the TRIMD.TIS bit and EESIPR.TWBIP bit to 1 to generate an interrupt request.

TFE Bit (Transmit Frame Error)

When the TFE bit is 1, it indicates that one or more of the TFS bits is 1.

TFP[1:0] Bits (Transmit Frame Position)

The TFP[1:0] bits indicate which part of a transmit frame corresponds to the transmit buffer indicated by this descriptor. The TFP[1:0] and TD1.TBL bit settings must be logically consistent in the previous and next descriptors.

TACT Bit (Transmit Descriptor Valid)

The TACT bit indicates that this descriptor is valid. The TACT bit is set to 1 by software. This bit becomes 0 when the transmit frame has been transferred or when the transmission is aborted.

(2) Transmit Descriptor 1 (TD1)

Bit	Symbol	Bit Name	Description	R/W
b15 to b0	—	Reserved	The read value is 0. The write value should be 0.	R/W
b31 to b16	TBL	Transmit Buffer Length	Set a valid byte length of the corresponding transmit buffer. Set a value equal to or greater than 1.	R/W

TD1 is used to set a valid byte length of the transmit buffer.

(3) Transmit Descriptor 2 (TD2)

Bit	Symbol	Bit Name	Description	R/W
b31 to b0	TBA	Transmit Buffer Address	Set the start address of the transmit buffer. When the TD1.TBL bit value is 1 to 16 bytes, align it on a 32-byte boundary.	R/W

TD2 is used to set the start address of the transmit buffer.

38.3.1.2 Receive Descriptor

Figure 38.4 shows the relation between a receive descriptor and receive buffer. The receive frame and receive buffer configuration can be selected from one buffer per frame (single-buffer frame transmission) or multiple buffers per frame (multi-buffer frame transmission) by setting the receive descriptor. If the receive buffer length (RBL) is set to 0, operation indicated by the descriptor is not guaranteed.

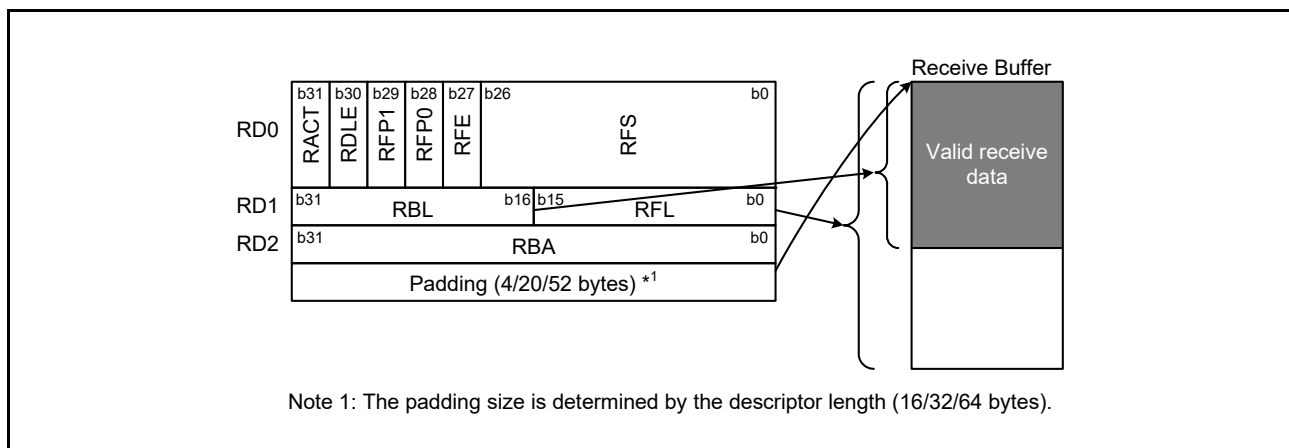


Figure 38.4 Relation between Receive Descriptor and Receive Buffer

(1) Receive Descriptor 0 (RD0)

Bit	Symbol	Bit Name	Description	R/W
<u>b26 to b0</u>	<u>RFS</u>	Receive Frame Status	Set all bits to 0 when creating a descriptor. After write-back, the bits indicate the following: [for EDMACn] RFS26 to RFS10: Reserved RFS9: Receive FIFO overflow (the value is equivalent to the EESR.RFOF flag) RFS8: Receive abort is detected (the value is equivalent to the EESR.RABT flag). RFS7: Multicast address frame is received (the value is equivalent to the EESR.RMAF flag). RFS6 and RFS5: Reserved RFS4: Alignment error is detected (the value is equivalent to the EESR.RRF flag). RFS3: Frame-too-long error (the value is equivalent to the EESR.RTLF flag) RFS2: Frame-too-short error (the value is equivalent to the EESR.RTSF flag) RFS1: PHY-LSI receive error (the value is equivalent to the EESR.PRE flag) RFS0: CRC error (the value is equivalent to the EESR.CERF flag) When each bit becomes 1, it indicates that the corresponding error has occurred during frame reception. When any of the RFS bits becomes 1, the RFE bit also becomes 1 (set the TRSCER register to select whether bits RFS7 and RFS4 are reflected in the RFE bit). When any bit from RFS3 to RFS0 becomes 1, the RFS8 bit also becomes 1. [for the PTPEDMAC] RFS26 to RFS10: Reserved RFS9: Receive FIFO overflow (the value is equivalent to the EESR.RFOF flag) RFS8: Reserved RFS7: Receive port (the value is equivalent to the EESR.RPORT flag) RFS4: PTPV2 packet is received (the value is equivalent to the EESR.PVER flag). The PTPEDMAC can receive only PTP packets. If a non-PTP packet is received, the packet is not transferred to the PTPEDMAC, and it is discarded. RFS3 to RFS0: Type of the received PTP message (the value is equivalent to the EESR.TYPE[3:0] flags) Each bit indicates the status of the received frame.	R/W
<u>b27</u>	<u>RFE</u>	Receive Frame Error	[for EDMACn] 0: No error has occurred in the received frame. 1: An error has occurred in the received frame. [for the PTPEDMAC] Reserved	R/W
<u>b29, b28</u>	<u>RFP[1:0]</u>	Receive Frame Position	b29 b28 0 0: Receive buffer indicated by this descriptor is the middle of a receive frame (frame information is incomplete). 0 1: Receive buffer indicated by this descriptor is the end of a receive frame (frame information is complete). 1 0: Receive buffer indicated by this descriptor is the head of a receive frame (frame information is incomplete). 1 1: Receive buffer indicated by this descriptor is all of a receive frame (one buffer per frame).	R/W
b30	RDLE	Receive Descriptor List End	When this bit is 1, it indicates that this descriptor is the last one of the descriptor list.	R/W
<u>b31</u>	<u>RACT</u>	Receive Descriptor Valid	Indicates that this descriptor is valid.	R/W

Note: Bits for write-back are underlined.

RD0 indicates the receive frame status.

RFE Bit (Receive Frame Error)

When the RFE bit is 1, it indicates that one or more of the RFS bits is 1 (set the TRSCER register to select whether bits RFS7 and RFS4 of EDMACn are reflected in the RFE bit).

RFP[1:0] Bits (Receive Frame Position)

The RFP[1:0] bits indicate which part of a receive frame corresponds to the receive buffer indicated by this descriptor.

RACT Bit (Receive Descriptor Valid)

The RACT bit indicates that this descriptor is valid. The RACT bit is set to 1 by software. This bit becomes 0 when all data has been transferred to the receive buffer indicated by RD2 or when the receive buffer becomes full.

(2) Receive Descriptor 1 (RD1)

Bit	Symbol	Bit Name	Description	R/W
<u>b15 to b0</u>	<u>RFL</u>	Receive Frame Length	These bits indicate the length (number of bytes) of the receive frame stored in the buffer. The number of bytes for padding set by the RPADIR register is not included. These bits are written back to the descriptor corresponding to the end of a frame.	R/W
b31 to b16	RBL	Receive Buffer Length	These bits indicate the byte length of the corresponding receive buffer. Set an integral multiple of 32 as the buffer length.	R/W

Note: Bits for write-back are underlined.

RD1 indicates the receive buffer length. When reception is completed, the receive frame length is written back.

(3) Receive Descriptor 2 (RD2)

Bit	Symbol	Bit Name	Description	R/W
b31 to b0	RBA	Receive Buffer Address	These bits indicate the start address of the receive buffer. Align the buffer address on a 32-byte boundary.	R/W

RD2 indicates the start address of the receive buffer.

38.3.2 Transmission

When setting the EDTRR.TR bit to 1 while the ETHERCn.ECMR.TE bit is 1, the EDMAC reads the descriptor following the previously used descriptor in the transmit descriptor list (or the descriptor indicated by the TDLAR register after a reset). When the TACT bit is 1 in the transmit descriptor (TD0), the EDMAC sequentially reads transmit data from the start address of the transmit buffer indicated by transmit descriptor 2 (TD2) and transfers them to the ETHERC via the transmit FIFO. The ETHERC creates a transmit frame and starts transmission to the MII/RMII. When all data indicated by the TD1.TBL bit is transferred, write-back is performed according to the TD0.TFP[1:0] bits as follows:

- When the TD0.TFP[1:0] bits are 00b or 10b (frame information is incomplete), the TD0.TACT bit is written back.
- When the TD0.TFP[1:0] bits are 01b or 11b (frame information is complete), bits TD0.TACT, TD0.TFS, and TD0.TFE are written back.

When the TD0.TACT bit in the read descriptor is 1, the EDMAC continues reading descriptors and transmit frames.

When the TD0.TACT bit in the read descriptor is 0, the EDMAC sets the EDTRR.TR bit to 0 and stops transmission.

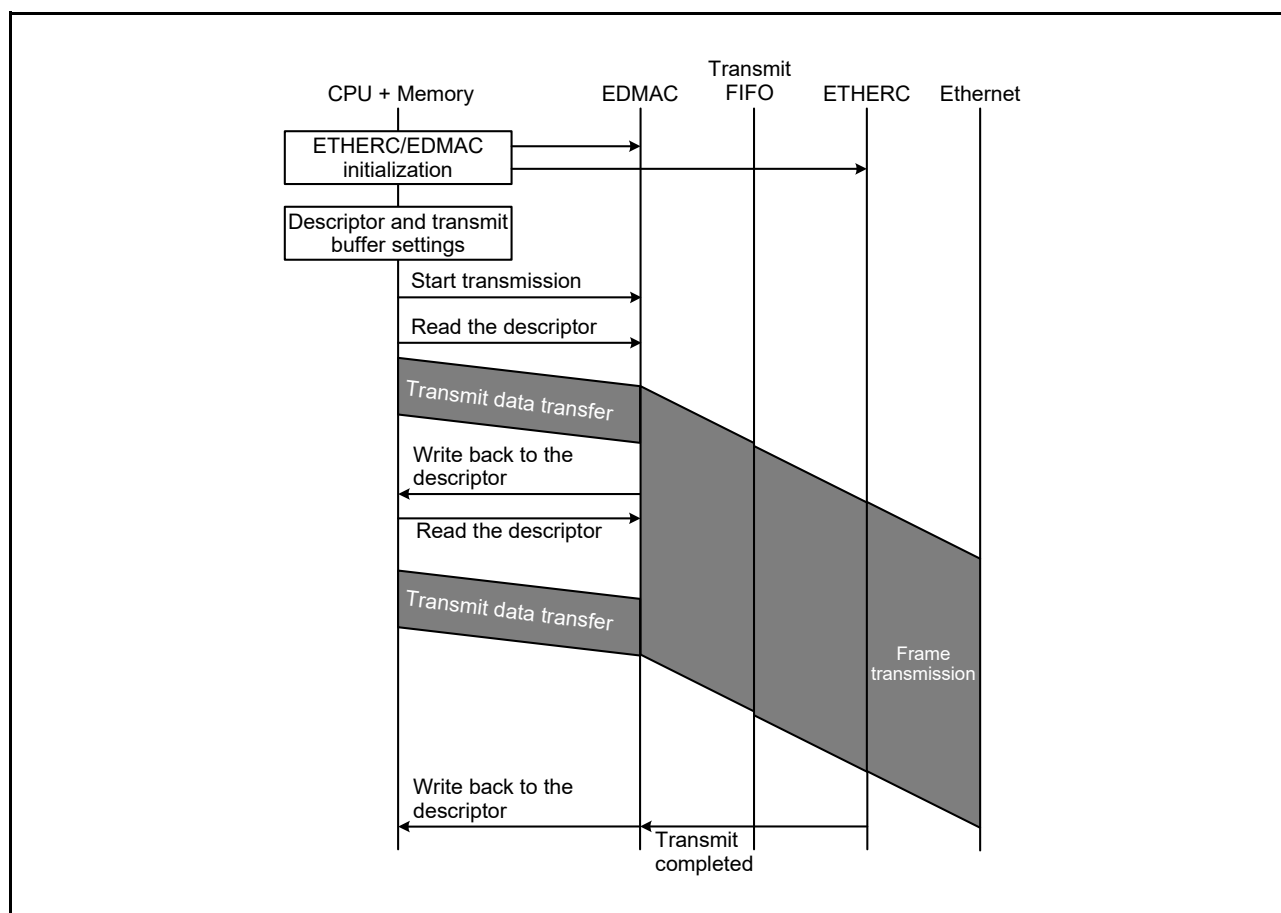


Figure 38.5 Example of Transmission Flow

38.3.3 Reception

When setting the EDRRR.RR bit to 1 while the ETHERCn.ECMR.RE bit is 1, the EDMAC reads the receive descriptor following the previously used descriptor (or descriptor indicated by the RDLAR register after a reset) and then waits for reception. While the RD0.RACT bit is 1, if the data stored to the receive FIFO is 32 bytes or more, or if the end byte of the frame is stored in the receive buffer, the EDMAC transfers data from the receive FIFO to the receive buffer indicated by receive descriptor 2 (RD2). If the data length of the received frame is longer than the buffer length set by the RBL bit in receive descriptor 1 (RD1), the EDMAC writes back 10b or 00b to the RD0.RFP[1:0] bits and 0 to the RD0.RACT bit when the receive buffer becomes full, and then the EDMAC reads the next data. After that, the EDMAC transfers data to another receive buffer. When the frame reception is completed or when the frame reception is aborted by an error, the EDMAC writes back 11b or 01b to the RD0.RFP[1:0] bits, 0 to the RD0.RACT bit, and the receive frame length to the RD1.RFL bit*1. When the RMCR.RNR bit is 1, the EDMAC reads the next descriptor and waits for reception. When the RNR bit is 0, the EDMAC sets the EDRRR.RR bit to 0 and stops reception.

Note 1. When the frame reception is aborted if the data stored in the receive FIFO is less than 16 bytes, the data is not transferred, but discarded. In this case, the corresponding error flag does not become 1.

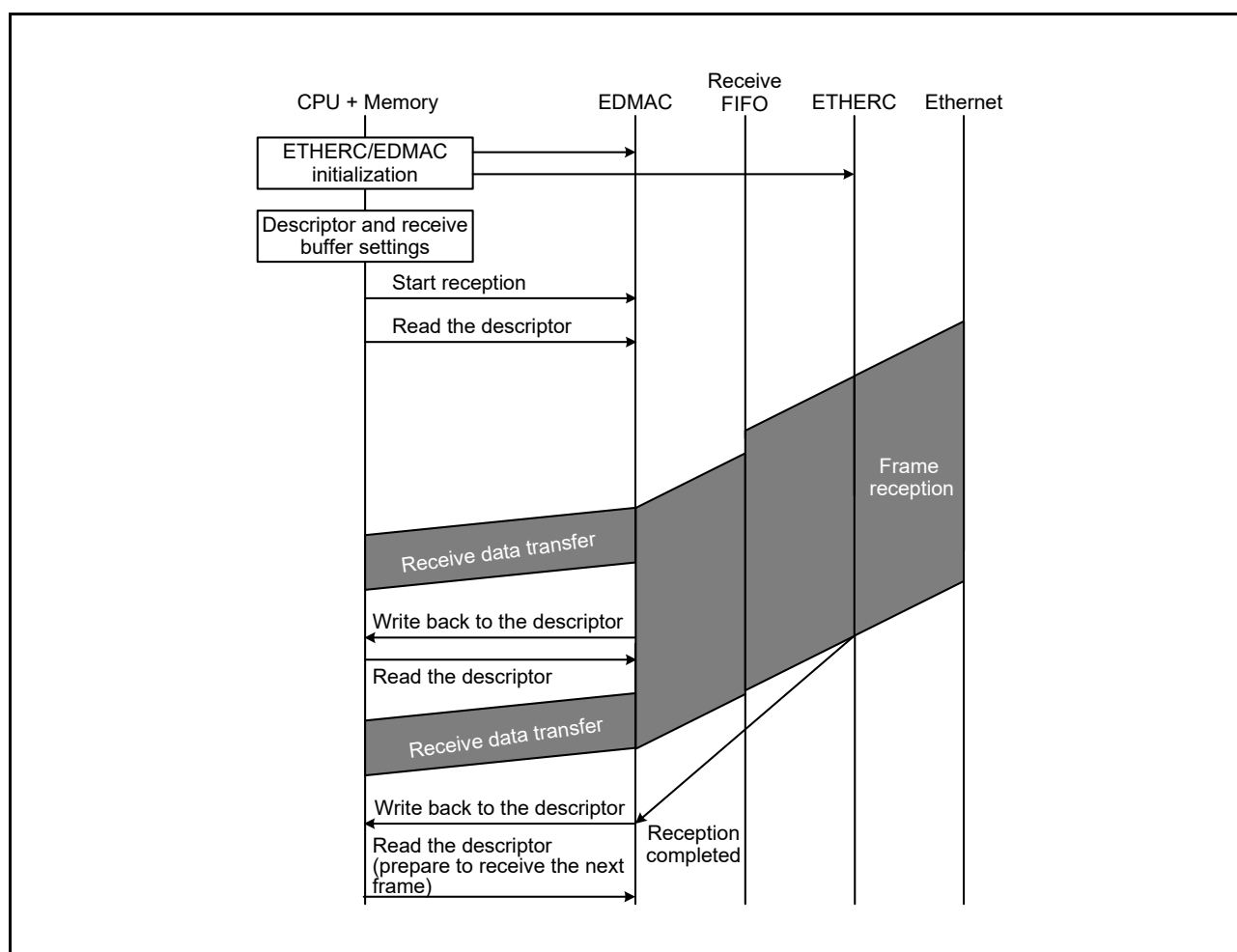


Figure 38.6 Example of Reception Flow

38.3.4 Multi-Buffer Frame Transmission

38.3.4.1 Error Processing While Transmitting a Multi-Buffer Frame

If an error occurs during multi-buffer frame transmission, the EDMAC performs the processing shown in Figure 38.7. In the figure, when the TACT bit of transmit descriptor 0 (TD0) is 0, the descriptor indicates that all data in the buffer has been successfully transmitted. When the TACT bit is 1, the descriptor indicates that data in the buffer has not yet been transmitted. If a frame transmit error *1 occurs in the head or middle of the frame while the TD0.TACT bit is 1, the EDMAC stops data transmission from the transmit FIFO and EDMAC data transfer, and sets the TD0.TACT bit to 0. After that, the EDMAC reads the next descriptor to see if the descriptor indicates the middle of the frame (TD0.TFP[1:0] bits are 00b) or the end of the frame (TD0.TFP[1:0] bits are 01b). When the descriptor indicates the middle of the frame, the EDMAC sets the TD0.TACT bit to 0 and reads the next descriptor. When the descriptor indicates the end of the frame, the EDMAC not only sets the TD0.TACT bit to 0, but also writes back to bits TD0.TFE and TD0.TFS. After an error occurs, data in the buffer is not transmitted until write-back to the descriptor for the end of the frame. When the corresponding transmit error interrupt is enabled in the EESIPR register, an interrupt request is generated immediately after write-back to the descriptor for the end of the frame.

Note 1. For EDMACn, a transmit retry-over condition, late collision, or loss of carrier is detected, or a carrier is not detected. For the PTPEDMAC, the MAC address does not match the set value.

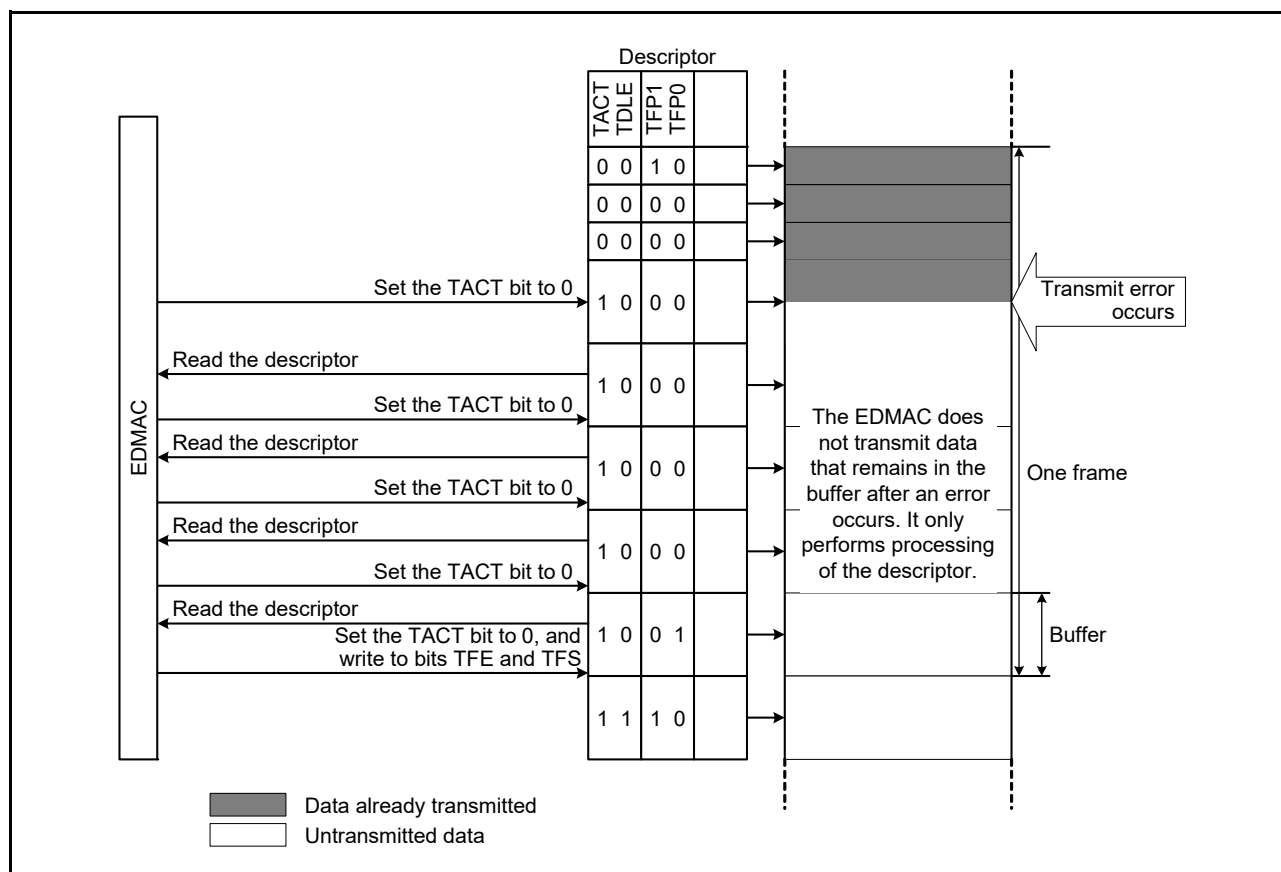


Figure 38.7 EDMAC Operation After a Transmit Error Occurs

38.3.4.2 Error Processing While Receiving a Multi-Buffer Frame

If an error occurs during multi-buffer frame reception, the EDMAC performs the processing shown in Figure 38.8.

In the figure, when the RACT bit of receive descriptor 0 (RD0) is 0, the descriptor indicates that data has been successfully received in the buffer. When the RACT bit is 1, the descriptor indicates that data has not yet been received in the buffer. If a frame receive error ^{*1} occurs, the EDMAC stops receiving data of the frame, but it transfers data that has already been stored in the receive FIFO to the receive buffer.

When the receive buffer becomes full during transfer, the EDMAC sets the RACT bit to 0 and the RFP[1:0] bits to 10b or 00b and reads the next descriptor. After all data in the receive FIFO has been transferred, the EDMAC writes back the status to the descriptor.

When the corresponding receive error interrupt is enabled in the EESIPR register, an interrupt request is generated immediately after write-back to the descriptor. When there is a request to receive a new frame, the EDMAC continues reception using the descriptor following the descriptor where the error occurred.

Note 1. For EDMACn, a CRC error, PHY-LSI receive error, frame-too-short error, frame-too-long error, or alignment error is detected. For the PTPEDMAC, a parity error is detected.

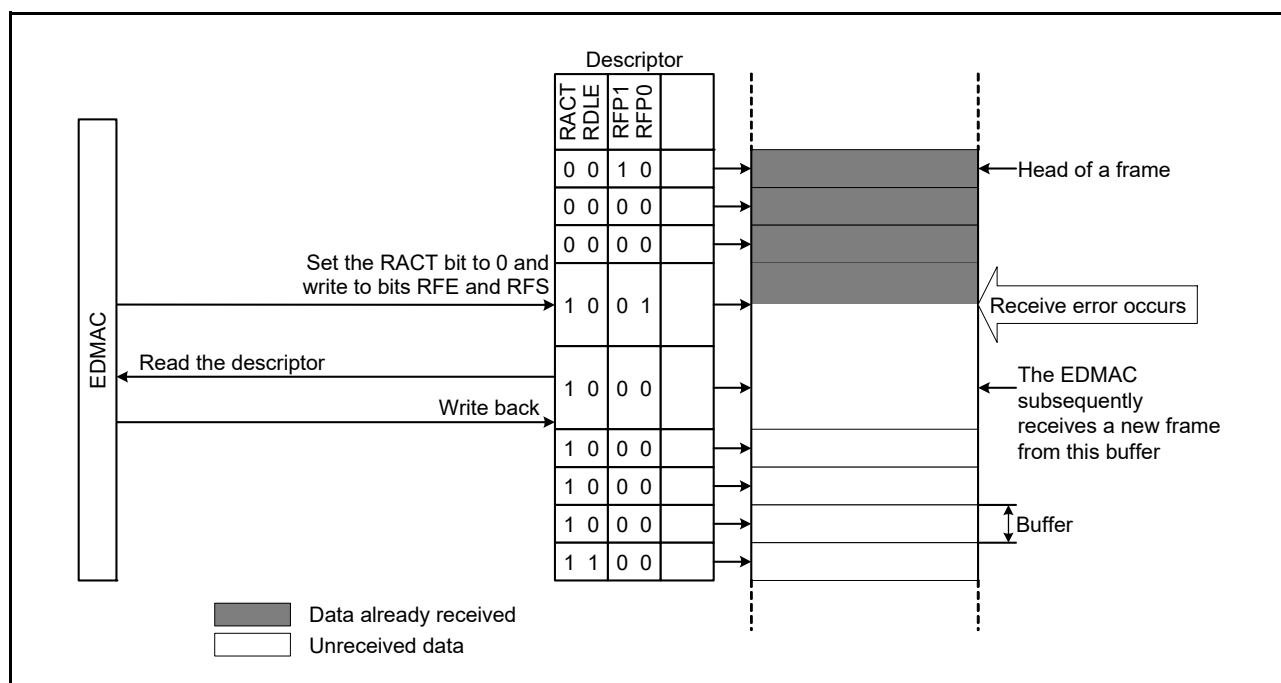


Figure 38.8 EDMAC Operation After a Receive Error Occurs

38.3.5 EDMAC Channel Priority

Priority of the three EDMAC channels (EDMAC0, EDMAC1, PTPEDMAC) is determined in a round-robin fashion. Each time transfer of one channel is completed, the channel becomes the lowest priority. This operation is shown in Figure 38.9. After a reset, the priority is EDMAC0 > EDMAC1 > PTPEDMAC.

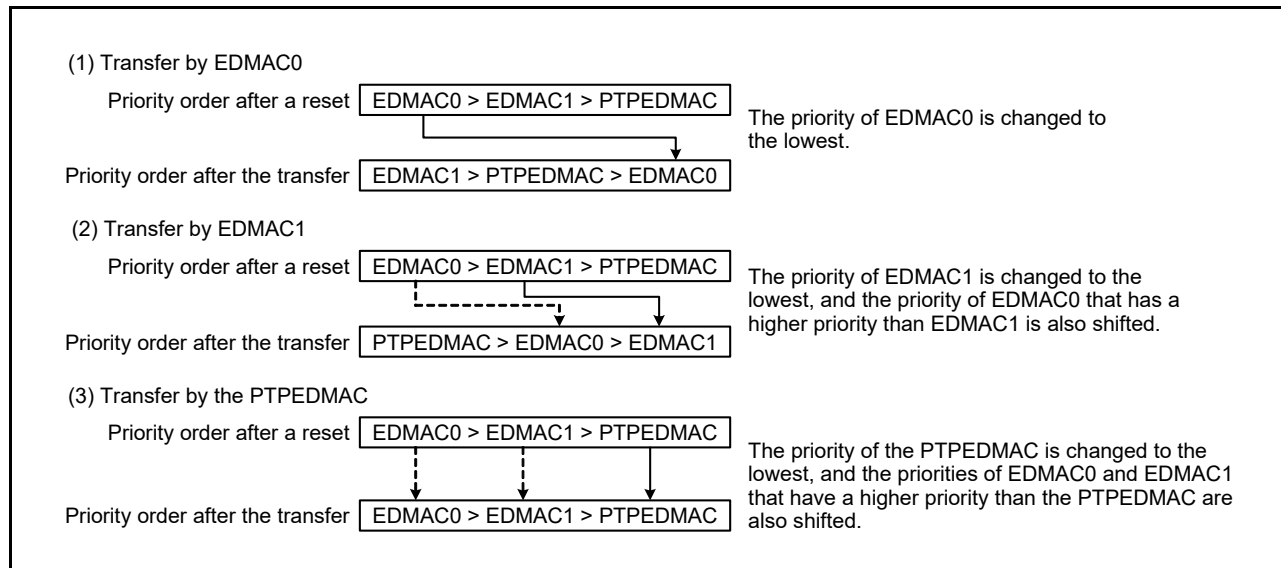


Figure 38.9 Operation in Round-Robin Fashion

Figure 38.10 shows change of the channel priority order when transfer requests are concurrently generated to EDMAC0 and the PTPEDMAC and a transfer request to the EDMAC1 while EDMAC0 is transferring data.

1. Transfer requests are concurrently generated to EDMAC0 and the PTPEDMAC.
2. Since EDMAC0 has a higher priority than the PTPEDMAC, EDMAC0 starts a transfer (PTPEDMAC waits to transfer).
3. A transfer request is generated to EDMAC1 during the transfer by EDMAC0 (EDMAC1 and the PTPEDMAC wait to transfer).
4. After EDMAC0 ends the transfer, the priority of EDMAC0 is changed to the lowest.
5. Since EDMAC1 has higher priority than the PTPEDMAC at this time, EDMAC1 starts a transfer (PTPEDMAC waits to transfer).
6. After EDMAC1 ends the transfer, the priority of EDMAC1 is changed to the lowest.
7. The PTPEDMAC starts a transfer.
8. After the PTPEDMAC ends the transfer, the priority of PTPEDMAC is changed to the lowest.

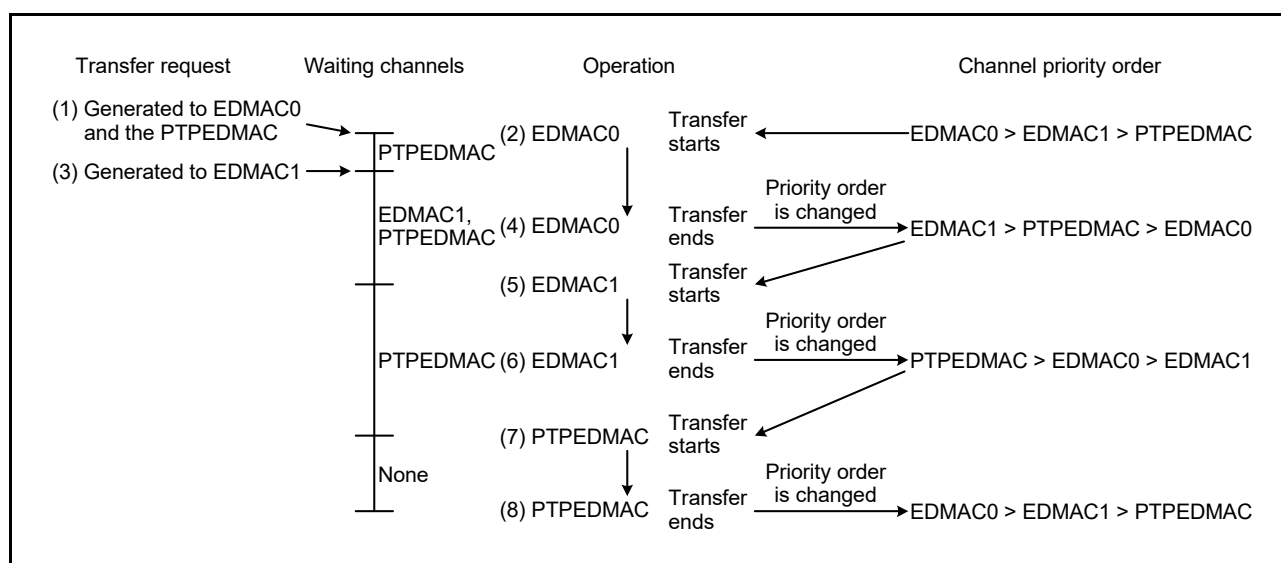


Figure 38.10 Channel Priority in Round-Robin Fashion

38.4 Interrupts

When any of the status flags in the EESR register becomes 1 while the corresponding interrupt request enable bit in the EESIPR register is 1, EDMACn outputs an EINTn interrupt request or the PTPEDMAC outputs a PINT interrupt request to the CPU.

38.5 Usage Notes

38.5.1 Setting the Module-Stop Function

Module stop control register B (MSTPCRB) is used to enable/disable the EDMAC operation. After a reset, the EDMAC is stopped. After exiting the module-stop state, registers can be accessed. Refer to section 11, Low Power Consumption for details on the module-stop function.

38.5.2 Stopping the EDMAC during Operations

When stopping EDMAC operation by using a WAIT instruction or module-stop function while the EDMAC is running, confirm that bits EDTRR.TR and EDRRR.RR are 0. If the EDMAC is stopped while the EDTRR.TR or EDRRR.RR bit 1, the data for the frame that is being transmitted or received may not be complete, and EDMAC operation after exiting the sleep mode or module-stop state is not guaranteed.

38.5.3 Illegal Address Access Detection

When the illegal address access detection on the EDMAC operation is to be used, set the BSC.BEREN.IGAEN bit and ICU.IER02.IEN0 bit to 1.

If the BSC.BERSR1.MST[2:0] bits are 110b when a bus error interrupt occurs, set the EDMR.SWR bit to 1 (EDMAC and ETHERC are reset) and then configure the EDMAC and ETHERC again.

38.5.4 Note on Software Reset

If the EDMR.SWR bit is set to 1 while the EDMAC is running, data in the address range from 0000 0000h to 0000 001Fh may be destroyed. Do not use the address range from 0000 0000h to 0000 001Fh when using the Ethernet controller.